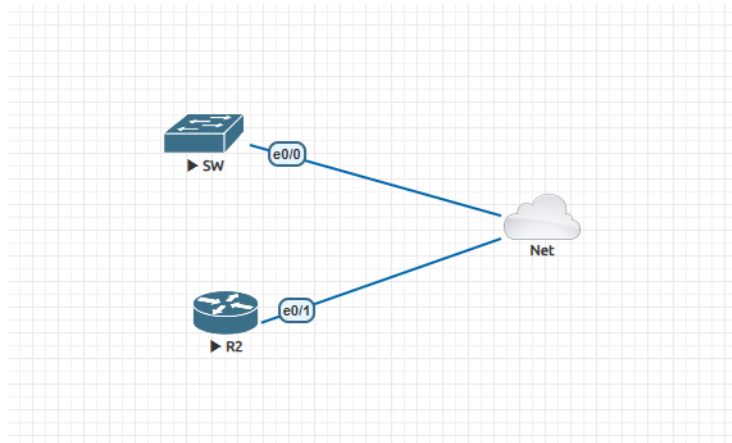


TP 3 : Exercice 2 | R4.09 DevOps

Infrastructure:



Le Cloud « Net » est connecté directement à la VM via une interface réseau privée hôte.

Adresse IP de l'infrastructure:

VM ansible : 192.168.56.103

ROUTEUR : 192.168.56.102

Switch (vlan 1) : 192.168.56.104

Configuration pour pouvoir accéder aux équipements réseau :

conf t

hostname R1 (ou Sw1)

ip domain-name local

username Routeur (ou Switch) privilege 15 secret toto

enable secret enable-pw

crypto key generate rsa

ip ssh version 2

line vty 0 4

login local

transport input ssh

exit

Connexion depuis la VM Ansible :

Pour se connecter au :

- **Switch** : `ssh Switch@192.168.56.104 -o HostKeyAlgorithms=+ssh-rsa`
- **Routeur** : `ssh Routeur@192.168.56.102 -o HostKeyAlgorithms=+ssh-rsa`

Question 4)

Afin de pouvoir faire fonctionner notre playbook et récupérer notre fichier backup, on doit créer le dossier via cette commande :

mkdir backups

Et pour lancer la configuration :

`ansible-playbook -i hosts.yaml backup_router.yaml`

```
root@R409: /home/vboxuser/Documents/TP3/ansible-TP3# ansible-playbook -i hosts.yaml backup_router.
yaml
PLAY [AUTOMATIC BACKUP OF RUNNING-CONFIG] *****

TASK [DISPLAYING THE RUNNING-CONFIG] *****
[WARNING]: ansible-pylibssh not installed, falling back to paramiko
[WARNING]: Deprecation warnings can be disabled by setting `deprecation_warnings=False` in ansibl
e.cfg.
[DEPRECATION WARNING]: The paramiko connection plugin is deprecated. This feature will be removed
from ansible-core version 2.21.
[DEPRECATION WARNING]: Passing `warnings` to `exit_json` or `fail_json` is deprecated. This featu
re will be removed from ansible-core version 2.23. Use `AnsibleModule.warn` instead.
ok: [R1]

TASK [SAVE OUTPUT TO ./backups/] *****
changed: [R1]

PLAY RECAP *****
R1                : ok=2    changed=1    unreachable=0    failed=0    skipped=0    rescu
ed=0    ignored=0
root@R409: /home/vboxuser/Documents/TP3/ansible-TP3#
```

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Résultat du playbook :

```
root@R409:/home/vboxuser/Documents/TP3/ansible-TP3# ls -l backups/
total 4
-rw-r--r-- 1 root root 1194 Apr  6 21:20 show_run_R1.txt
root@R409:/home/vboxuser/Documents/TP3/ansible-TP3# cat backups/show_run_SWITCH.txt
cat: backups/show_run_SWITCH.txt: No such file or directory
root@R409:/home/vboxuser/Documents/TP3/ansible-TP3# cat backups/show_run_R1.txt
Building configuration...

Current configuration : 1134 bytes
!
! Last configuration change at 23:13:05 EET Mon Apr 6 2026 by Switch
!
version 15.4
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
!
hostname ROUTEUR
```

Question 5)

Création du dossier pour la récupération des configurations :

```
mkdir ios_show
```

Lancement du playbook :

```
ansible-playbook -i hosts.yaml router_ipv4_config.yaml
```

Résultat du playbook :

```
admin_sw.yaml  backup_router.yaml  backups  hosts.yaml  ios_show  router_ipv4_config.yaml
vboxuser@R409:~/Documents/TP3/ansible-TP3$ cat ios_show/show_ip_intr_br_R1.txt
Interface                IP-Address      OK? Method Status Protocol
Ethernet0/0              unassigned      YES unset  administratively down down
Ethernet0/1              192.168.56.102  YES DHCP    up        up
Ethernet0/2              10.10.10.1      YES manual  up        up
Ethernet0/3              unassigned      YES unset  administratively down downvboxuser@R409:~/Documents/TP3/v
```

Question 6)

Modification des hosts pour ajouter le switch :

```
GNU nano 7.2                                hosts.yaml
router:
  hosts:
    R1:
      ansible_host: 192.168.56.102
      ansible_network_os: cisco.ios.ios
      ansible_user: Routeur
      ansible_password: toto
      ansible_become: yes
      ansible_become_method: enable
      ansible_become_password: toto

    SW1:
      ansible_host: 192.168.56.104
      ansible_network_os: cisco.ios.ios
      ansible_user: Switch
      ansible_password: toto
      ansible_become: yes
      ansible_become_method: enable
      ansible_become_password: toto
```

Lancement du playbook :

ansible-playbook -i hosts.yaml admin_sw.yaml

Résultat de la commande :

```
root@R409: /home/vboxuser/Documents/TP3/ansible-TP3# ansible-playbook -i hosts.yaml admin_sw.yaml

PLAY [ADMINISTRATION DU SWITCH] *****

TASK [CHANGE HOSTNAME ET CREER VLAN 200] *****
[WARNING]: ansible-pylibssh not installed, falling back to paramiko
[WARNING]: To ensure idempotency and correct diff the input configuration lines should be similar
to how they appear if present in the running configuration on device
[WARNING]: Deprecation warnings can be disabled by setting `deprecation_warnings=False` in ansible
e.cfg.
[DEPRECATION WARNING]: The paramiko connection plugin is deprecated. This feature will be removed
from ansible-core version 2.21.
[DEPRECATION WARNING]: Passing `warnings` to `exit_json` or `fail_json` is deprecated. This featu
re will be removed from ansible-core version 2.23. Use `AnsibleModule.warn` instead.
changed: [SW1]

TASK [CONFIGURE INTERFACES E0/1 ET E0/2 DANS VLAN 200] *****
changed: [SW1] => (item=Ethernet0/1)
changed: [SW1] => (item=Ethernet0/2)

PLAY RECAP *****
SW1                : ok=2    changed=2    unreachable=0    failed=0    skipped=0    rescu
ed=0    ignored=0
```

Résultat sur le switch, avec le changement de nom, création du VLAN200 et les interfaces :

```
Connection to 192.168.56.104 closed.  
boxuser@R409:~$ ssh Switch@192.168.56.104 -o HostKeyAlgorithms=+ssh-rsa  
Switch@192.168.56.104) Password:  
  
SWITCH#en  
SW_LV#  
  
SW_LV#sh vl br  


| VLAN | Name               | Status    | Ports        |
|------|--------------------|-----------|--------------|
| 1    | default            | active    | Et0/0, Et0/3 |
| 200  | VLAN0200           | active    | Et0/1, Et0/2 |
| 1002 | fddi-default       | act/unsup |              |
| 1003 | token-ring-default | act/unsup |              |
| 1004 | fddinet-default    | act/unsup |              |
| 1005 | trnet-default      | act/unsup |              |

  
SW_LV#  
  
interface Ethernet0/0  
switchport mode access  
!  
interface Ethernet0/1  
switchport access vlan 200  
switchport mode access  
!  
interface Ethernet0/2  
switchport access vlan 200  
switchport mode access  
!  
interface Ethernet0/3  
!  
interface Vlan1  
ip address 192.168.56.104 255.255.255.0
```

(Tous les fichiers seront disponibles dans le fichier compressé dans le dépôt)